

Towards Multimodal VR Trainer of Voice Emission and Public Speaking -Work-in-Progress

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Figure 1: Project logo and example screen from the location designed for training a remote public speech.

ABSTRACT

GlossoVR is a virtual reality (VR) application that combines training in public speaking in front of a virtual audience and in voice emission in relaxation exercises. It is accompanied by digital signal processing (DSP) and artificial intelligence (AI) modules which provide automatic feedback on the vocal performance as well as the behavior and psychophysiology of the user. In particular, we address parameters of speech emotions, prosody and timbre, and the user's hand gestures and eye movement. The prototype is in the proof of concept phase, and we are developing it in accordance with the user-centered design paradigm. In this article reports the work in progress, focusing on the approaches, datasets and algorithms applied in the current state of the *glossoVR* project.

Index Terms: Virtual Reality—Speech—Visualization—Voice analysis; Affective computing—Gestures—Biometry—Eye tracking

1 INTRODUCTION

This article reports the progress of the *glossoVR* project, which we first introduced in 2020 [15]. It consists of two modes: training oral

presentation in front of a virtual audience in the presentation mode and training voice and speech in the relaxation mode. Since that stage, we expanded the number of analyzed modalities as well as the range of parameters within each modality. We also report on the current state of human factors research in the context of *glossoVR*, especially considering the COVID-19 pandemic, which resulted in the increased ratio of remote speeches versus traditional in-person meetings.

1.1 Motivation

The necessity of speaking in public, especially delivering oral presentations, accompanies many roles and professions, starting with schools and universities, talking with clients or reporting to a project team, and ending with jobs that require professional voice emissions, like journalism, acting or teaching. At the same time, vocal tract diseases due to improper voice effort are among the main occupational diseases [32]. Also, voice timbre has become a part of a person's image, and voice quality can affect a person's quality of life [27]. Conversely, experiencing constant stress from public speaking, stage fright or glossophobia can lead to a poorer quality of life. The *Public Speaking Anxiety* (PSA) and glossophobia, a fear of speaking in public, have been studied i.a., for students in Malaysia [42], Ghana [8], Pakistan [31] and Spain [24], which would suggest that the problem itself is intercultural.

Currently, the possibility of using machine learning with the use

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of big data, access to wearable sensors, and high-quality graphics real-time rendering of immersive virtual environments, are the key elements allowing for the creation of a VR application that supports voice and speech training with personalized feedback and progress monitoring. Controlling the stimuli and features of the training environment corresponds with method applied in Virtual Reality Therapy, Virtual Reality Immersion Therapy, Simulation For Therapy, Virtual Reality Exposure Therapy and Computerized Cognitive Behavior Therapy.

1.2 Current State-of-the-Art in VR Speech Training

The first attempts at using immersive environments to treat social phobia [19] and glossophobia occurred about two decades ago [30], [35]. Presently, among commercial applications, the most popular approaches are Ovation [1] and Virtual Speech [44].

Some newer, enhanced approaches also incorporated biometric modalities like heart rate monitoring while speech practicing [13] or analyzing eye gaze [10]. Another direction in the development of VR trainers is developing responsive virtual audiences [10] or creating population-specific models [46].

As is pointed out in the recent study [6], validating this kind of behavioral therapy requires a rigorous methodology and carefully drafted experimental protocol. Recently, new investigations into the acceptance and performance of the VR public speaking training [28] were also published. Moreover, some other works observed that the impact of the virtual audience could also depend on its quality, e.g. the level of realism of the virtual humans [38].

The authors of [33] investigated the effectiveness of VR for the training of teachers' vocal skills by simulating a lesson in a realistic VR environment characterized by adjustable constraints such as background noise and fidgety children. The results showed that teachers significantly increased their voice frequency, intensity, and intonation and made longer pauses while speaking *in vivo* and *in vitro*, compared to the control condition. The possibility of use in the therapy of stuttering has also been analyzed [21]. The possible problem is, however, whether users will feel comfortable speaking in the VR headset at all, which we have explored in an experimental environment [4].

The *glossVR* aims to provide a complex approach to voice and speech training by supplementing the public presentation mode with a set of relaxation exercises and speech emission training. Such an approach is helpful in order to reduce stress and prepare the vocal tract for speech production. The complexity of the presented technique is reflected in a number of parameters and modalities monitored by the system, which provides the user with detailed feedback on their behavior.

In contrast to currently existing solutions, e.g., VirtualSpeech [44], that relies to a large degree on textual feedback, we plan to employ VR-based visual analytics [40] to report results to the user. The visual analytics agenda is comprised of at least these two main components: (1) interactive visualizations and (2) analytical reasoning about the data [41]. The focus of this approach is to allow the user to gain new and deeper insights into the data rather than the speed of analysis itself [40]. This approach seems well-suited to the needs of *glossVR* users by focusing on individual user progress and speech skills improvement.

What is more, the training mode will be supplemented with built-in gamification elements to visualize progress and support developing proper habits for everyday voice emission.

On the basis of the users' needs and wants, as well as several voice training and public speaking experts' recommendations, we have designed a solution which could support different phases of voice performance coaching. Apart from the standard mode of training public speaking in front of an audience, we introduce voice emission and relaxation modes in natural environments. An example of visualizing pitch contour in real-time is presented in Fig. 2). We added

these new features in order to train voice emission with selected recommended methods in the form of exercises and games, e.g., blooming a meadow of flowers or making trees grow by maintaining proper vocalization.

The prototype is being developed with the use of Unity, with a strong focus on compliance with Meta's and OpenXR's standards to ensure the broadest possible range of supported VR/XR devices.

2 HUMAN FACTORS RESEARCH

We apply user-centered design process, which has been successful in the recent approaches to using VR for Exposure Therapy [26]. A mixture of quantitative and qualitative methods were applied to investigate the problems speakers meet during preparation for public speaking. In 2019, an online survey was conducted on a sample of 386 respondents, out of which 67% were female, and 85% of respondents were within the 18-35 age range. We use the results to explore the attitude towards public speaking (e.g., emotions, physiological reactions, ways of managing stress, perceived difficulty with various types of public speaking), previous experiences, both positive and negative, sources of improving competencies in this field and amount of time allocated for preparation. We found out that 2/3 of young adults experience only negative emotions in the context of public speaking. We are currently performing the survey's next iteration to verify if the COVID-19 pandemic changed the public speaking context in terms of public speaking anxiety.

The goal of this ongoing qualitative research is to explore how users deal with the situation of public speaking. We also wanted to identify problems they may face while using various speech training applications. We performed two batches of tests: with twelve respondents before the pandemic in 2019 and with another twelve respondents in 2022. The findings seemed to point out the importance of adding supplementary locations which would simulate giving a remote speech instead of a virtual audience in the same room (Fig. 1). The results also suggest the need to design separate paths for such remote speech training (the difference in recommendations between in-person and remote speech effect, e.g. usage of gestures or gaze).

3 NEW SPEECH CORPORA

We are in the process of collecting new speech corpora, which will be used to model speech patterns during public presentations. These speech corpora are:

- (A) A corpus of typical public speech samples. It is being collected from media as a sample from six categories of public speech: TED talks, lectures from Massive Open Online Courses (e.g. Coursera), vlogs, podcasts, investor pitches, and politicians' speeches. Each category is represented by at least 50 recordings of various speakers, e.g., 50% female and 50% male, with each recording lasting three minutes, except the investors' pitches which are sometimes shorter. The purpose of this corpus is to calculate statistical models of speech parameters that are typical for each genre.
- (B) A corpus of excellent public speech samples. In this approach, we are collecting the best speeches using crowdsourcing. We do this thanks to an Internet survey designed to acquire links to exemplary public presentations and the respondents' opinions on the presentation and the speaker.

The purpose of the first corpus (A) is to calculate statistical models of speech parameters in the typical sample of each genre, whilst we plan to use the second corpus (B) to model the expected acoustic parameters of public speech.

Additionally, in order to model each speech feature, we use further speech corpora designed to analyze specific speaker traits and state features (e.g. emotions, timbre, speech rate).

4 SPEAKER BEHAVIOR ANALYSES

In this section, we present modalities selected to be monitored in the *glossoVR* engine. They were selected on the basis of the criteria of importance for VR speech training defined together with the speech trainer, as well as the technical feasibility study.

4.1 Voice Analysis

Apart from semantic information, various vocal traits impact a public speech's performance and overall impression. In particular, in the *glossoVR* project, we aim to analyze speech in the contexts of many parameters such as emotions, speech rate, intonation contours, stress detection, the occurrence of pauses, and speech type. Besides efficient detection and analysis of voice traits, we also tackle the problem of selecting the most effective method for presenting the outcomes of speech analysis to the user in VR. We plan to utilize the immersive visual analytics [40] approach as the basis of our work in this domain.

Emotional Vocalization. Although keeping the actual desired emotional state might be challenging, a professional speaker should be able to control the emotions that are conveyed to an audience. For example, speakers who sound rude will not succeed in listeners' engagement. In one of our previous studies, we investigated acoustic correlates of polite and impolite speech [12], and we found increased speech rate and voice modulation when a speaker tends to be polite. Multiple approaches focus on the classification of emotions from speech. In the *glossoVR* project, the state-of-the-art algorithm that employs Wav2vec 2.0 [2, 5, 29] structure is fine-tuned on the Polish Emotional Speech Database [14] and used as the baseline system. In particular, due to the limited number of emotions present in the publicly available datasets of speech patterns, we focused on the detection of boundaries between the reference neutral state and negative emotions such as anger and sadness as the ones which affect most the impression of speech.

Stress detection. Regardless of their actual emotional and cognitive state, speakers typically aim not to be perceived as too stressed during their presentation. Hence, in our system, we work towards including a module which detects the stressed voice. To establish a baseline detection reference level, we recorded a database of students presenting for a large university audience. Our preliminary studies using 529 short audio recordings showed that the correlation between labels annotated in a 5-level Likert scale [23] by six experts is equal to 0.4. For evaluation, we extracted 96 samples for which annotators scored the recordings as equal or less than 2 for neutral and equal or more than 4 for stressed with a standard deviation equal to or less than 1. Using a combination of various publicly available datasets¹ for emotion recognition with fear emotion labelled as stress and training the binary stress classifier with GMM-based modelling of *eGeMAPSv02* feature set [11], extracted with *OpenSMILE*², we obtained the balanced accuracy of stress detection at the level of approx. 60%, which matches the prior state-of-the-art results [17].

Speech Rate Analysis. Measuring the rate at which words are exclaimed impacts not only the intelligibility of a speaker but also their perceived politeness [36]. Hence the tempo should be adapted to the audience and topic of the presentation. In order to measure this speech parameter, we use the *Temporal Correlation and Spectral Sub-band Correlation* (TCSSBC) [45] method as it provides high speech rate estimation accuracy while leveraging low computational complexity. It has been empirically proven that with carefully adapted hyperparameters, such as amplitude and time thresholds, the algorithm estimates four syllables per second for natural speech, which indicates a correct calibration with the ground truth [34].

¹In specific, we used *EMO-DB*, *TESS*, *RAVDESS*, *MESD*, *JL* corpus reported in <https://superkogito.github.io/SER-datasets/>

²<https://github.com/audereing/opensmile-python>

Intonation. Proper voice modulation forms the speech melody and hence may affect the attentiveness of the audience. Alternatively, when a speaker does not have proper control of their speech, public speaking might be considered either boring or funny. The *glossoVR* system provides a speech melody evaluation method based on the detection and classification of pitch contours. To fulfill a rigorous performance requirement, we used for pitch estimation the probabilistic *YIN* (*pYIN*) [25]. We also perform the real-time classification into *typical*, *low*, and *highly variable* classes using gender-dependent *Gaussian Mixtures Models* (GMM) obtained from the pitch contours in subsequent time-frames with a frame-wise subtracted mean. Additionally, we relied on intonation contours to serve as a basic parameter for speech emotion recognition [9].

Pauses. The pauses are one of the speech prosody features which should be analyzed statistically in long analysis windows [16]. Therefore, descriptors of the number and length of pauses are usually given in about 1 min frames. Filled pauses (i.e., fillers) are among the most annoying for the audience speech influences [3]. They can be typical for the language, but also a personal manner of speech and the majority of them can be easily detected [47].

Speech Type/Genre. We analyzed the corpus of typical examples of 6 speech genres in order to find out if there are acoustic parameters which are significantly different among speech categories. Classification methods were applied in order to measure the probability of the speech genres being automatically distinguishable. The highest accuracy was obtained for the Random Forest classifier, which recognized six classes with 52% accuracy. The most distinctive categories were TED talks and vlogs (>90% accuracy).

4.2 Gestures Analysis

Focus on recognizing gesture quality instead of recognizing specific gestures (e.g. sign language items, interaction gestures) is a new approach not previously met in typical gesture recognition in VR. In this case, we define gesture quality as the style of gesticulation performed during the speech, which includes dynamics and range. On the basis of a prepared database of gestures recorded with the VR headset and controllers positioning, three aspects of gestures: (1) dynamics, (2) range, and (3) distance of hands from each other were assessed perceptually by ten judges on the basis of video recording. Among the judges, there were experts (e.g., psychologists and speech trainers) and non-experts. Each dimension was assessed on a scale of three values, i.e., low, medium and high. In the comparison of classifiers, the best results were achieved for the *Support Vector Machine* (SVM), where the classification accuracy was at the level of 75% for each dimension.

4.3 Eye-Tracking Analysis

Proper eye contact of the speaker with the audience lies among the essential aspects of interaction during public speaking [22]. The VR public speaking trainers are limited by the availability of headsets supporting eye-tracking and their lack of precision in measurement. While many popular solutions use gyroscopic measurements, new solutions with built-in eye-trackers are appearing (e.g., *HP Reverb Omnicept*, *Meta Quest Pro*). In eye-tracking data visualization, it is essential to find a balance in the displayed data in order to provide users with information without overwhelming or distracting them during a speech. In our implementation, it will be important to focus on measuring the frequency and duration of fixations, which will allow for the automation of feedback on users' eye movements during speeches and training. This warrants more conscious eye management to add emphasis to the content of the speech.

4.4 Psychophysiology Analysis

Simulations in virtual environments, especially when dealing with such an emotional situation as public speaking, will affect the signals from the body of users in many ways. Not only psychological



Figure 2: White contour visualizes voice pitch in real-time during intonation practicing in one of the relaxation modes. Visual feedback strengthens the consciousness of voice modulation and its stability over time.

factors and current internal processes of a human but also the characteristics of the experience and the type of task performed can influence the measured signals. We rely on a three-system model of stressful situations [18], (initially [20]), in which anxiety is felt at the cognitive (e.g., *I am a terrible speaker, I am sure I am wrong*), physiological (e.g., elevated heart rate) and behavioral level (e.g., avoiding eye contact with the public). The criteria that we adapt for selecting a suitable indicator are signal purity, ability to read and interpret the signal in real-life, resistance to user movement and ease of presenting the signal to the user in an interpretive/readable way. In the process of designing the best setup for *glossoVR*, we have started from the setup described in [37], which used *biosignals plus* sensors and *Pupil labs* eye-tracker, and currently, we are benchmarking wearable sensors as well as built-in sensors in selected VR headsets, e.g. *HP Reverb Omnicept*.

5 RESULTS REPORTING

We plan to capture an abundance of information from the analysis of human voice and behavior while performing a public speech. These data can have the form of categorical (e.g., speakers style or intention) or numerical information (e.g., speech pauses count and lengths) or can be represented by a time series (e.g., speech rate, voice vibration or modulation). Moreover, we can obtain additional information from the statistical analysis of such speech parameters through the lenses of affective computing, such as stress detection [7] or reasoning about the speaker's concealed emotions [43]. Some feedback will be provided in real-time, and the rest will be shown retrospectively as a report after the presentation.

As there is a range of varying information that *glossoVR* returns to its user to provide precise, individual feedback, there is also a plethora of visualization techniques that can be applied here. However, we have to carefully select a subset of these methods to provide a solution beneficial to a potentially large body of different users with varying prior exposure to speech training and/or VR systems. Moreover, the visual analytics solution should be designed in such a way as to focus the users' attention on essential results rather than flooding them with unstructured information. Hence, to ascertain and select the appropriate visualization features as well as interaction technique [39, 40], we will design a series of controlled user experiments utilizing a mixed-methods approach that was previously successfully deployed in VR-based visual analytics [39, 40].

6 CONCLUSION

Public speaking and voice emission are complex processes which can be analyzed on many levels and from many perspectives. The assessment rules cannot be easily generalized because of the multimodality of communication in both the verbal and non-verbal

domains. Moreover, other factors that had to be considered here are large interpersonal variability, as well as the subjective nature of speaker assessment. In our approach, several modalities are combined to provide measurable feedback that is as rich as possible to either the user or their trainer, allowing them to monitor the progress of training.

ACKNOWLEDGMENTS

This work was supported by program Lider grant no 0230/L-11/2019 by the National Center for Research and Development, Poland. Special thanks to Mrs Anna Sowa-Paciorek for her support in the administration of this project.

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